

# ToughTorq Reaction Washer System - Technical Data

Reaction-arm-free controlled bolting with a matched coaxial dual-socket system

## Reaction Washer Pair

Design principle: Wedge-locking anti-loosening system designed in accordance with DIN 25201 principles

Material: 1.1191 (C45E)

Coating: Black zinc flake (flZnnc)

Hardness: 485 ´ – 25 HV 0.3

Corrosion resistance: Minimum 1,000 hours NSS testing to ISO 9227

Metric size range: M16â M48

Imperial size range: 5/8â 1-3/4 in

Fastener strength: bolts through class 12.9; nuts through class 12

## System Function

The gear-shaped lower washer provides the reaction interface at the joint.

Torque is contained coaxially through the matched dual socket rather than an external reaction arm.

The wedge-locking washer pair is intended to resist vibration-induced self-loosening.

# ToughTorq Reaction Washer System - Selection & Compatibility

Select every interface as one complete bolting system

## Matched Dual Socket

Coaxial socket system that engages the fastener and the gear-shaped Reaction Washer profile so reaction torque is contained at the joint.

Matched configurations available for common powered torque tools

Match the dual socket to the selected Reaction Washer size and torque-tool interface.

Common electric, pneumatic, and hydraulic torque tools can be supported with matched configurations.

## Passive-Side Option

Friction-based backup washer for through-bolt connections where the passive-side nut or bolt head cannot be physically counter-held.

Wave-shaped locking ribs on both faces resist passive-side rotation during tightening

Use where the passive-side nut or bolt head cannot be physically counter-held.

## Selection Checks

Confirm fastener diameter and Reaction Washer nominal size.

Confirm torque-tool drive/interface and the matching dual-socket configuration.

Confirm available axial and radial clearance around the joint.

Confirm material/coating requirements for the application environment.

Additional nominal sizes, materials, and coatings are available on request.